Dong-Hwan Jang

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Research Interests

Generative Modeling & Visual Understanding
Image and Video Generation, Diffusion Models, World Models, Vision-Language Models (VLMs),
3D Geometric Consistency, Physical Plausibility, Inference-Time Alignment, Model Merging.

Education

University of Illinois Urbana-Champaign 2025 – Present
Ph.D. Student in Computer Science. *Advisor: Ismini Lourentzou*

Seoul National University 2020 – 2023
M.S. in Electrical and Computer Engineering. *Advisor: Bohyung Han*

Carnegie Mellon University Fall 2022
Visiting Scholar, Korean Government-funded AI project program.

Seoul National University 2013 – 2020
B.S. in Electrical and Computer Engineering; B.S. in Technology Management.
Department of Liberal Studies. *Summa Cum Laude; ranked first in class.*

Current Research

JOLT: Geometry-Consistent Video Generation
Led problem formulation, method design, implementation, experiments, and manuscript writing. Developed training-free video steering with uncertainty-selective 3D Gaussian perturbation ensembles and Feynman–Kac sampling to assess reconstruction stability during generation. *In progress.*

SALM: Sim-Anchored Latent Matching
Proposing a video similarity metric using counterfactual simulation supervision to separate physical dynamics from appearance changes, with planned use in reranking outputs of frozen video generators.

Publications

[1] **JOLT: Uncertainty-Selective Self-Ensembling for Geometric Consistency in Video Generation**
Dong-Hwan Jang, Ying Shen, Adheesh Sunil Juvekar, Onkar Kishor Susladkar, and Ismini Lourentzou.
In progress

[2] **RewardFlow: Generate Images by Optimizing What You Reward**
Onkar Kishor Susladkar, **Dong-Hwan Jang**, Tushar Prakash, Adheesh Sunil Juvekar, Vedant Shah, Ayush Barik, Nabeel Bashir, Muntasir Wahed, Ritish Shrirao, and Ismini Lourentzou.
CVPR 2026

[3] **PyraTok: Language-Aligned Pyramidal Tokenizer for Video Understanding and Generation**
Onkar Kishor Susladkar, Tushar Prakash, Adheesh Sunil Juvekar, Kiet A. Nguyen, **Dong-Hwan Jang**, Inderjit S. Dhillon, and Ismini Lourentzou.
CVPR 2026

[4] **Model Stock: All We Need is just a few Fine-Tuned Models**
Dong-Hwan Jang, Sangdoo Yun, and Dongyoon Han.
ECCV 2024 **Oral** (**Top 2.3%** among submitted papers)

[5] **Merge and Bound: Direct Manipulations on Weights for Class Incremental Learning**
Taehoon Kim, **Dong-Hwan Jang**, and Bohyung Han.
CVPR Workshop on Continual Learning in Computer Vision 2024

[6] **Pooling Revisited: Your Receptive Field is Suboptimal**
Dong-Hwan Jang, Sanghyeok Chu, Joonhyuk Kim, and Bohyung Han.
CVPR 2022

[7] **DS4C Patient Policy Province Dataset: a Comprehensive COVID-19 Dataset for Causal and Epidemiological Analysis**
Jimi Kim*, Seojin Jang*, Woncheol Lee*, Joong Kun Lee*, and **Dong-Hwan Jang***.
NeurIPS Workshop 2020 (* Equal contribution.)

Research & Industry Experience

University of Illinois Urbana-Champaign — *Research Assistant* Summer 2026 – Present
Advisor: Ismini Lourentzou

- Led design, implementation, and controlled evaluation of a pre-execution LLM code reviewer to detect metric fabrication, data leakage, invalid checkpoints, and task-constraint violations.
- Improved an existing LangGraph prototype with retry memory, cross-milestone code sharing, checkpoint compatibility, and numeric checks, enabling end-to-end ML-claim evaluation.

Samsung AI Center, Korea — *Research Engineer* Jan 2024 – Aug 2025

- Built a wafer-inspection anomaly detection framework using distribution-shift modeling and semi-supervised learning, improving reliability and reducing false positives.
- Researched fine-tuning methods for semiconductor manufacturing and deployed results to production systems.

NAVER AI Cloud, Korea — *Research Intern* Jul 2023 – Dec 2023

- Proposed Model Stock for robust fine-tuning through weight merging; published at ECCV 2024 as an oral paper.
- Advised by Dongyoon Han and Sangdoo Yun.

Mind's Lab, Korea — *Research Scientist* Feb 2019 – Sep 2019

- Researched weakly supervised segmentation and spatially adaptive colorization; fulfilled South Korean military service.

TNC Technology, Korea — *Software Engineer* Oct 2016 – Feb 2019

- Designed and deployed a Java payment gateway for KOCES, a Korean payment processor; fulfilled South Korean military service.

Technical Experience

- **RewardFlow (CVPR 2026):** Developed an inversion-free, reward-guided Langevin framework for diffusion-based image editing. Designed prompt- and region-aware rewards for controllable edits without latent inversion or additional fine-tuning.
- **Model Stock (ECCV 2024 Oral):** Proposed geometry-based, layer-wise merging of just two fine-tuned CLIP models, outperforming Model Soup on in-distribution and out-of-distribution benchmarks.
- **DynOPool (CVPR 2022):** Developed differentiable pooling to optimize receptive fields, achieving up to 5.6 percentage points higher accuracy on fine-grained recognition and approximately 33% fewer MACs on ImageNet with comparable accuracy.
- **Implicit Neural Representation for Motion Deblur:** Designed spatially variant image deblurring using implicit representations. *U.S. Patent Application No. 17/973,809.*
- **DS4C:** Curated patient- and policy-level COVID-19 data for epidemiological modeling, forecasting, and causal analysis.

Scholarships & Awards

- **2025–2029:** Kwanjeong Educational Foundation Study Abroad Scholarship; USD 25,000 per year for Ph.D. study.
- **2023–2024:** Korean Government Scholarship for Overseas Study; one of 56 selected candidates across all disciplines.
- **2022:** OnDream Global Scholarship Award, Hyundai Chung Mong-Koo Foundation; recognized the CVPR 2022 paper.
- **2021–2022:** OnDream Future Technology Scholarship, Hyundai Chung Mong-Koo Foundation; full tuition and support.
- **2015–2016:** National Scholarship for Science and Engineering, Korea Student Aid Foundation; full tuition.

Invited Talks & Presentations

- **Korean Conference on Computer Vision (2022):** "Pooling Revisited: Your Receptive Field is Suboptimal." Invited oral delivered by Prof. Bohyung Han on my behalf; poster presented personally.
- **Databricks (2020):** One-hour invited talk as part of the DS4C team, "The Complexities around COVID-19 Data."

Teaching Experience

- **Fall 2020:** Teaching Assistant, *430.329: Introduction to Algorithms*, Seoul National University.
- **July 2019:** Teaching Assistant, *Samsung AI Expert Course*, Seoul National University.
- **Jan 2019:** Teaching Assistant, *Hyundai Motors AI Expert Course*, Seoul National University.

Academic Service

Reviewer: CVPR 2023, 2026; NeurIPS 2022, 2023, 2026; TMLR.

Technical Skills

Languages: Python, C/C++, Java. **ML Frameworks:** PyTorch, Hugging Face Transformers.
Tools: Git, Docker, Linux, W&B, TensorBoard, LaTeX. **Training:** Mixed precision (AMP).